

API_REFERENCE.md

VSSC Infrastructure Management Portal

Backend API Reference & Integration Guide

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PURPOSE OF THIS DOCUMENT

This document serves as the definitive API reference for the VSSC Infrastructure Management Portal.

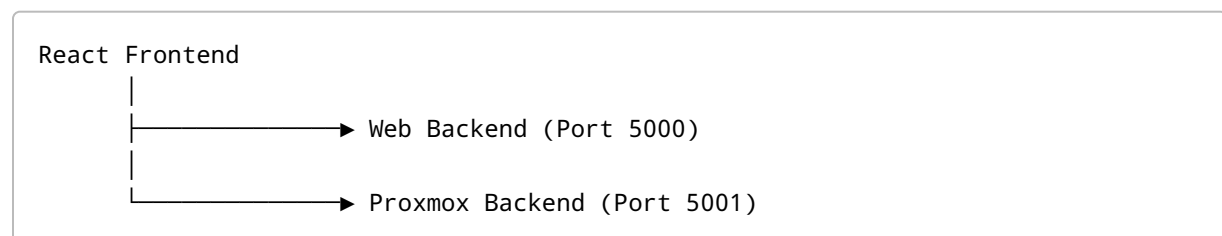
It documents:

- Available endpoints
- Request methods
- Request payloads
- Response structures
- Authentication requirements
- Database dependencies
- Frontend integration guidance
- Error handling recommendations

This document should be consulted before implementing or modifying any frontend feature.

SYSTEM ARCHITECTURE

The application communicates with two backend services.



WEB BACKEND

Purpose

Handles:

- Authentication
- Session management
- User creation
- Password changes
- VM request workflows

Default Port:

5000

AUTHENTICATION APIS

POST /auth/login

Purpose

Authenticate a user.

Request

```
{
  "staff_code": "admin",
  "password": "password"
}
```

Success Response

```
{
  "message": "Login successful",
  "staff_code": "admin",
  "role": "admin"
}
```

Failure Response

```
{
  "error": "Invalid credentials"
}
```

Database

```
ccds_db.user_role
```

Frontend Usage

Login Page

GET /auth/check

Purpose

Validate current session.

Success Response

```
{
  "staff_code": "admin",
  "role": "admin"
}
```

Failure Response

```
{
  "error": "Not authenticated"
}
```

HTTP Status:

```
401
```

Frontend Usage

ProtectedRoute

Session Validation

Auto Logout

POST /auth/logout

Purpose

Destroy active session.

Response

```
{  
  "message": "Logged out successfully"  
}
```

Frontend Usage

Logout Button

Session Cleanup

POST /auth/add-user

Purpose

Create a new dashboard user.

Access

Admin Only

Request

```
{  
  "staff_code": "VS12345",  
  "role": "view_only"  
}
```

Response

```
{
  "message": "User created successfully",
  "default_password": "vssc@isroVS12345"
}
```

Database

user_role

Frontend Usage

Administration

User Management

POST /auth/change-password

Purpose

Change password.

Request

```
{
  "staff_code": "VS12345",
  "new_password": "newpass",
  "confirm_password": "newpass"
}
```

Response

```
{
  "message": "Password changed successfully"
}
```

Database

user_role

Frontend Usage

Settings

Password Page

STAFF DIRECTORY APIS

GET /staff/search

Purpose

Search employees.

Query Parameter

q=VS1

Response

```
[
  {
    "staff_code": "VS10106",
    "name": "AJITH MR",
    "entity": "MME",
    "division": "XX"
  }
]
```

Database

empdetails

Frontend Usage

Ownership Assignment

User Search

VM REQUEST APIS

GET /vms

Purpose

Retrieve VM request forms.

Response

List of requested VMs and assigned users.

Database

```
vms  
users  
vm_users
```

Frontend Usage

VM Request Dashboard

POST /vms

Purpose

Submit new VM request.

Request

VM specification object.

Includes:

- Name
- RAM
- CPU
- OS
- Storage
- Assigned Users

Response

```
{  
  "message": "VM added successfully"  
}
```

Database

```
vms  
users  
vm_users
```

Frontend Usage

Add VM Request

PROXMOX BACKEND

Purpose

Handles infrastructure synchronization and monitoring.

Default Port:

```
5001
```

CLUSTER APIS

POST /proxmox/cluster/sync

Purpose

Synchronize cluster data from Proxmox.

Response

```
{
  "message": "1 Clusters successfully synced"
}
```

Database

```
cluster
```

Frontend Usage

Sync Center

Notes

Critical operation.

Must remain functional.

GET /proxmox/cluster/

Purpose

Retrieve all clusters.

Response

```
[
  {
    "cluster_name": "fsgpucluster-25"
  }
]
```

Database

```
cluster
```

Frontend Usage

Dashboard

NODE APIS

POST /proxmox/nodes/sync

Purpose

Synchronize node information.

Response

```
{
  "message": "3 nodes successfully synced"
}
```

Database

```
node
```

Frontend Usage

Sync Center

GET /proxmox/nodes/

Purpose

Retrieve all nodes.

Response

```
[
  {
    "node_name": "fsgpu1",
    "ip": "10.41.32.2",
    "live_status": true,
  }
]
```

```
[
  {
    "total_mem": 123456,
    "total_cores": 64
  }
]
```

Database

node

Frontend Usage

Dashboard

Nodes Screen

Infrastructure Analytics

STORAGE APIS

POST /proxmox/storage/sync

Purpose

Synchronize storage information.

Response

```
{
  "message": "18 storages successfully synced"
}
```

Database

storage

Frontend Usage

Sync Center

GET /proxmox/storage/

Purpose

Retrieve storage resources.

Response

```
[
  {
    "storage_name": "pbs-backup",
    "storage_type": "NFS",
    "total_size": 20480
  }
]
```

Database

storage

Frontend Usage

Storage Dashboard

VM Details

Analytics

VM APIS

POST /proxmox/vms/sync

Purpose

Synchronize VM inventory.

Response

```
{
  "message": "80 VMs successfully synced"
}
```

Database

```
vm
vm_cluster_relation
vm_node_relation
vm_storage_relation
```

Frontend Usage

Sync Center

GET /proxmox/vms/vmData

Purpose

Retrieve VM inventory.

Response

```
[
  {
    "uuid": "...",
    "vm_name": "w22-ueid-cad",
    "vm_id": 115,
    "cpus": 4,
    "max_memory": 8192,
    "status": "running"
  }
]
```

Database

```
vm
vm_cluster_relation
vm_node_relation
```

```
cluster
node
```

Frontend Usage

VM Inventory

Dashboard

Analytics

GET /proxmox/vms/vmData/{uuid}

Purpose

Retrieve detailed VM profile.

Response

Includes:

- VM metadata
- Storage mappings
- Ownership mappings
- Personnel mappings

Database

```
vm
vm_storage_relation
vm_user_relation
storage
empdetails
```

Frontend Usage

VM Detail Drawer

VM Detail Page

OWNERSHIP APIS

POST /proxmox/vms/{uuid}/addUsers

Purpose

Assign users to VM.

Request

```
{
  "users": [
    {
      "staff_code": "VS10106"
    }
  ]
}
```

Response

```
{
  "message": "1 users added successfully"
}
```

Database

vm_user_relation

Frontend Usage

Ownership Assignment

Admin Functions

POST /proxmox/vms/{uuid}/removeUsers

Purpose

Remove VM ownership.

Request

```
{
  "staff_codes": [
    "VS10106"
  ]
}
```

Response

```
{
  "message": "Removed 1 users"
}
```

Database

vm_user_relation

Frontend Usage

Ownership Management

USER SEARCH APIS

GET /proxmox/users/search

Purpose

Search employee directory.

Query Parameter

query=VS1

Database

empdetails

Frontend Usage

Ownership Assignment

User Search

VISUALIZATION APIS

GET /proxmox/visualization

Purpose

Pre-aggregated chart data.

Response Structure

```
{
  "os": {},
  "gpu": {},
  "status": {},
  "node": {}
}
```

Database

```
vm
vm_node_relation
```

Frontend Usage

Dashboard

Analytics

Charts

REPORTING APIS

POST /proxmox/report

Purpose

Generate infrastructure reports.

Request

```
{
  "columns": [
    "vm_name",
    "ip",
    "cpus"
  ],
  "format": "csv"
}
```

Supported Formats

```
csv
pdf
xlsx
```

Database

```
vm
vm_node_relation
vm_storage_relation
vm_user_relation
storage
empdetails
```

Frontend Usage

Report Builder

Exports

MONITORING APIS

GET /api/nodes

Purpose

Retrieve node names.

Response

```
{
  "nodes": [
    "fsgpu1",
    "fsgpu2",
    "fsgpu3"
  ]
}
```

Frontend Usage

Monitoring Filters

GET /api/stall/{kind}

Purpose

Retrieve pressure metrics.

Supported Types

```
cpu
memory
io
```

Example

```
/api/stall/cpu
```

Query Parameters

```
nodes=fsgpu1,fsgpu2  
timeframe=hour  
topk=5
```

Response

```
{  
  "x": [],  
  "series": []  
}
```

Frontend Usage

Monitoring Dashboard

Pressure Analytics

Resource Trends

FRONTEND INTEGRATION RULES

Authentication

Always verify session before rendering protected routes.

Use:

```
GET /auth/check
```

Ownership Filtering

User dashboards must filter VMs using:

```
vm_user_relation
```

Never expose global VM inventory to normal users.

Sync Operations

Show:

- Loading state
- Success state
- Error state

Never leave sync operations silent.

Reports

Treat backend as source of truth.

Do not generate reports client-side.

ERROR HANDLING STRATEGY

For all API calls:

Handle:

400
401
403
404
500

Provide:

- User-friendly message
 - Retry option
 - Logging support
-

PERFORMANCE GUIDELINES

Cache where possible.

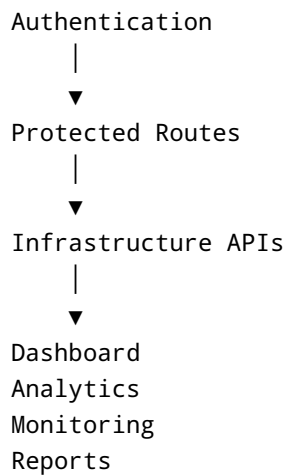
Avoid duplicate API calls.

Use loading skeletons.

Use lazy-loaded routes.

Paginate large datasets.

API DEPENDENCY MAP



FINAL API ASSESSMENT

Strengths:

- Clear separation of responsibilities
- Dedicated infrastructure backend
- Strong synchronization support
- Monitoring support
- Reporting support

Limitations:

- No audit API
- No historical metrics API
- Limited role granularity
- Session-based authentication

The current API surface is sufficient to build the planned Infrastructure Operations Portal without requiring backend modifications.